

Home

EDA dashboard

ML Models



EDA Dashboard

Air Quality · 13 Tabs

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Tab 5 — Feature Engineering



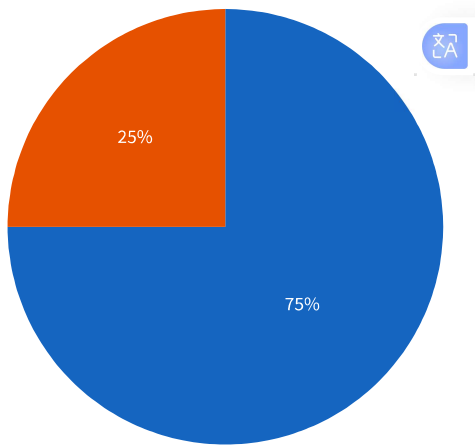
Time Features Created



Rush Hour Distribution

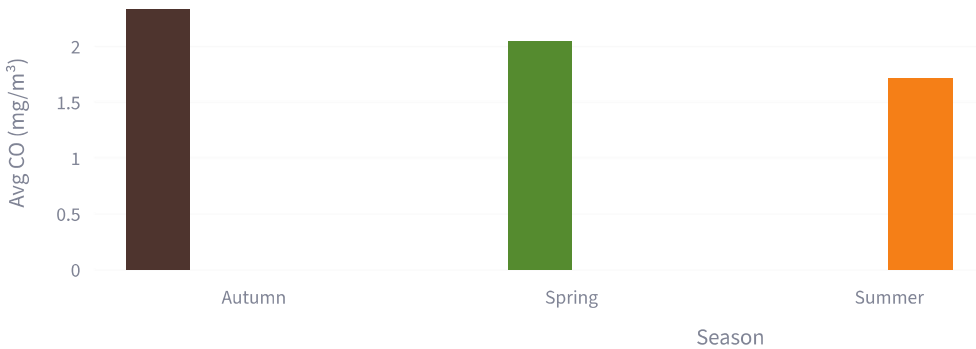
	Feature	Type	Description
0	Hour	int	0–23 hour of c
1	Day	int	Day of month
2	Month	int	1–12 month n
3	DayOfWeek	int	0=Mon ... 6=S
4	Weekend	binary	1 if Saturday c
5	Season	category	Spring/Summ
6	is_rush_hour	binary	1 if hour in [7,
7	high_pollution	binary (ML	1 if CO(GT) > n

Rush Hour vs Off-Peak Hours



Season Distribution

Average CO(GT) by Season



ML Target: high_pollution

High Pollution (1)

3,781

Low Pollution (0)

5,576

Balance

40.4% / 59...

✓ Rush hours represent ~25% of all records — a critical window for traffic-related pollution.

✓ Winter months show highest CO — cold air + heating combustion + less dispersion combine.

✓ high_pollution is nearly 50/50 balanced — no class_weight adjustment needed in ML models.

